

TB Disperser

DETAILED ENGLISH USER MANUAL

An all-pass phase-dispersion processor that reshapes transient arrival time without conventional EQ magnitude change, with a dedicated frequency LFO.



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Host-visible endpoints documented: 9

1. Product purpose

An all-pass phase-dispersion processor that reshapes transient arrival time without conventional EQ magnitude change, with a dedicated frequency LFO.

Sharp transients can need more weight or length without clipping, compression, or obvious tonal boost. TB Disperser rotates phase around a chosen frequency region so components arrive at different times, turning a short spike into a deeper sweep while largely preserving the static magnitude response.

What result it creates

- Longer and heavier attacks
- Frequency-focused phase sweeps
- Animated dispersion without DAW automation
- Transient design that differs from saturation or EQ

Signal flow

Input -> all-pass dispersion network shaped by Frequency, Pinch, Speed, and Amount -> optional frequency LFO -> output

2. Complete feature guide

Amount

Controls the total strength of phase rotation and therefore the audible length and curvature of the transient sweep.

Frequency

Sets the center or pivot of the dispersion. Low settings emphasize weight; higher settings create sharper zaps and metallic contours.

Pinch

Concentrates the phase action around the selected frequency. Higher values sound more focused and resonant.

Speed

Scales the time behavior of the dispersion independently of LFO Rate. Slower values stretch movement; faster values tighten it.

Frequency LFO

LFO Type selects Off, Sine, Triangle, Saw, Square, or Random. LFO Range sets frequency travel in octaves and LFO Rate controls motion speed.

Programs

Programs demonstrate tight phase shaping, drum bloom, bass bloom, metallic sweeps, pulse motion, and randomized scatter.

3. Quick start

1. Begin with LFO Off.

2. Set Frequency near the body of the transient.
3. Raise Amount until the attack clearly changes.
4. Use Pinch to focus the contour.
5. Use Speed to fit the source envelope.
6. Enable an LFO only when continuous motion is desired.

4. Practical workflows

Kick weight

1. Set Frequency low.
2. Use moderate Amount and low-to-medium Pinch.
3. Adjust Speed until the front edge becomes heavier without turning into a chirp.

Expected result: The kick feels deeper and longer without a conventional bass boost.

Animated synth zap

1. Set a mid or high Frequency.
2. Raise Pinch.
3. Choose Saw, Square, or Random LFO.
4. Set LFO Range and Rate for the rhythmic movement.

Expected result: The transient contour sweeps through the spectrum with repeatable motion.

5. Automation, gain staging, and session use

- Automate only controls that serve a clear musical transition. Fast movement of frequency, time, pitch, mode, oversampling, or algorithm selectors may intentionally create artifacts or require a host-safe transition.
- Match perceived loudness before judging quality. A louder setting will often appear better even when the processing decision is not better.
- Use the plug-in bypass endpoint for comparison and preserve an unprocessed source or earlier project version.
- Factory programs are starting points. Loading a program changes multiple parameters; save a new DAW preset after adapting it to the source.
- The parameter appendix is captured from the installed VST3 host contract. It lists every host-visible endpoint, its displayed range/options, default, automation status, and identifier.

6. Limits, cautions, and troubleshooting

- Phase dispersion changes waveform shape and peak timing even when the spectrum looks similar.
- Extreme settings can sound like pitch or resonant chirps.
- Recheck layered drums for cancellation.
- No audible change: confirm the plug-in and relevant sub-block are enabled, Mix/Amount is not at a neutral value, and the source contains energy in the processed region.
- Unexpected level: return input, output, send, feedback, and parallel-mix controls to conservative values, then rebuild the setting one block at a time.

- Automation does not match the panel: reload the project, confirm the DAW is addressing the current plug-in version, and check whether a factory program or state recall replaced values.
- Clicks or transitions: avoid sample-instant changes to algorithm, time, pitch, mode, or oversampling selectors unless the sound is intentional.

7. Complete host parameter reference

This appendix contains all 9 parameters exposed by the current installed VST3 to a host. Repeated EQ-band endpoints are intentionally listed rather than collapsed. Discrete options are printed in full when the host contract exposes them.

Parameter	Range / options	Default	Host status	Function
Amount VST3 ID 733630552	0.0 % to 100.0 %	70.0 %	Automatable	Sets how strongly the named process affects the signal.
Bypass VST3 ID 1652125811	Off, On	Off	Automatable; Bypass	Turns the complete plug-in processing path off or on through the host-visible bypass endpoint.
Frequency VST3 ID 2077459804	20.0 Hz to 20.00 kHz	200.0 Hz	Automatable	Sets the principal frequency, note, or spectral center used by this function.
LFO Range VST3 ID 685096968	±0.00 oct to ±4.00 oct	±1.00 oct	Automatable	Host-automatable control for LFO Range. Its full range and default are shown in this table; use it with the related feature section above.
LFO Rate VST3 ID 91373749	0.010 Hz to 10.00 Hz	0.250 Hz	Automatable	Sets the speed of modulation, movement, or repeated change.
LFO Type VST3 ID 91456271	OFF, SINE, TRI, SAW, SQR, RND	OFF	Automatable	Selects the operating algorithm, response shape, or tonal character for the named block.
Pinch VST3 ID 106671290	0.0 % to 100.0 %	0.0 %	Automatable	Host-automatable control for Pinch. Its full range and default are shown in this table; use it with the related feature section above.
Program VST3 ID 1886548852	Init, Tight Phase Shape, Drum Bloom, Slow Drum Zap, Slow Metallic Bloom, Bass Bloom, Synth Drift, Rising Zap, Two Point Pulse, Random Scatter	Init	Automatable	Selects a factory program. Loading a program recalls a coordinated set of plug-in parameters.
Speed VST3 ID 109641799	0.250 x to 2.000 x	2.000 x	Automatable	Host-automatable control for Speed. Its full range and default are shown in this table; use it with the related feature section above.

Documentation basis

Prepared from the current public catalog, current local product brief and implementation notes where available, existing English manuals where available, and a direct scan of the installed VST3 host contract. Internal implementation details that are not user-facing are intentionally excluded; all user-facing features and host-visible controls are documented.